COMMON STANDARD-OPEN NEIGHBORHOODS FOR TWO AFFINE OPENS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2206-2219, following the statement at 2198-2204.

Editorial note. The basis step 'We may choose an f in A, f notin p such that D(f) subset U cap V' uses that standard opens form a basis of Spec(A); the D(g) = D(g_A) step is the functoriality of standard opens under the affine morphism U -> V.

Allowed standard prerequisites (editorial).

- Standard opens refine affine intersections. lemma-standard-open (card L26395, part (2) and basis facts used at ~2210-2212). Chooses D(f) inside U cap V and notes standard opens of D(f) are standard opens of U.
- Standard opens are functorial for maps of affines. Algebra, Lemma algebra-lemma-spec-functorial (cross-chapter, used at ~2216-2217). D(g) in V pulled back to U equals D(g_A).

Lemma 0.1. *Let X be a scheme. Let U, V be affine opens of X, and let $x \in U \cap V$. There exists an affine open neighbourhood W of x such that W is a standard open of both U and V.*

Proof. Write $U = \mathrm{Spec}(A)$ and $V = \mathrm{Spec}(B)$. Say x corresponds to the prime $\mathfrak{p} \subset A$ and the prime $\mathfrak{q} \subset B$. We may choose an $f \in A$, $f \notin \mathfrak{p}$ such that $D(f) \subset U \cap V$. Note that any standard open of $D(f)$ is a standard open of $\mathrm{Spec}(A) = U$. Hence we may assume that $U \subset V$. In other words, now we may think of U as an affine open of V. Next we choose a $g \in B$, $g \notin \mathfrak{q}$ such that $D(g) \subset U$. In this case we see that $D(g) = D(g_A)$ where $g_A \in A$ denotes the image of g by the map $B \to A$. Thus the lemma is proved. □

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